部分实验数据分享

1. 电容实际值的测量

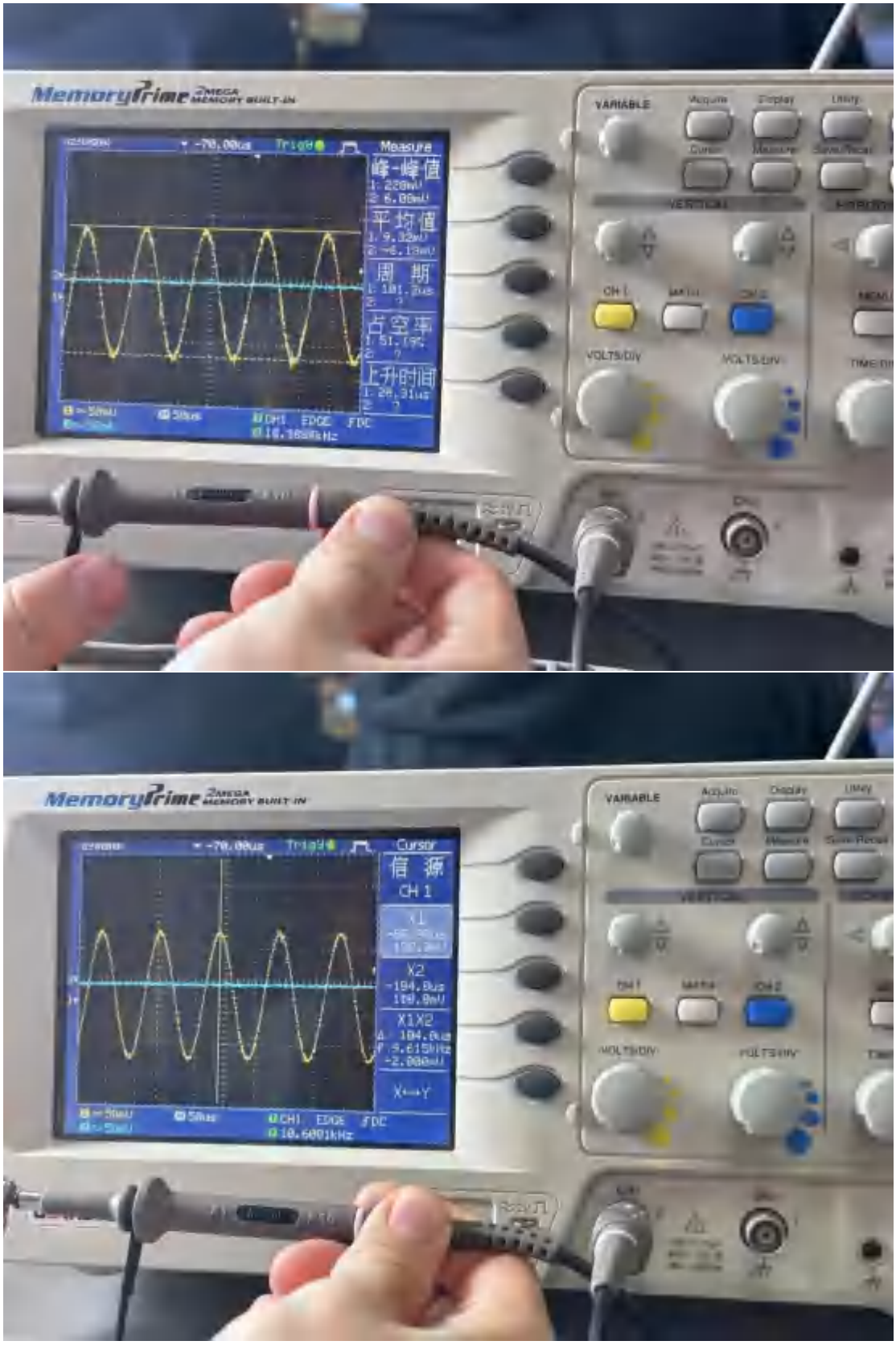


2. 电源使用联系—设定电源为 5V 与 12v 电流保持为 1A

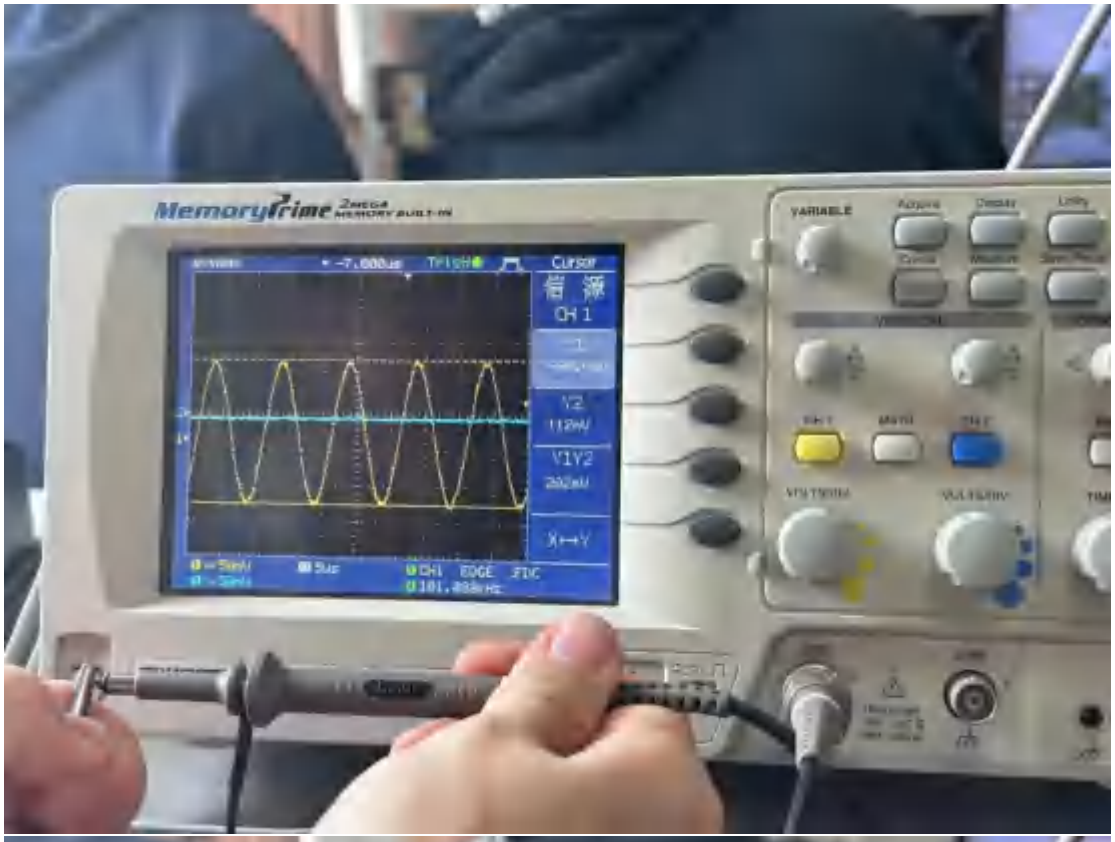


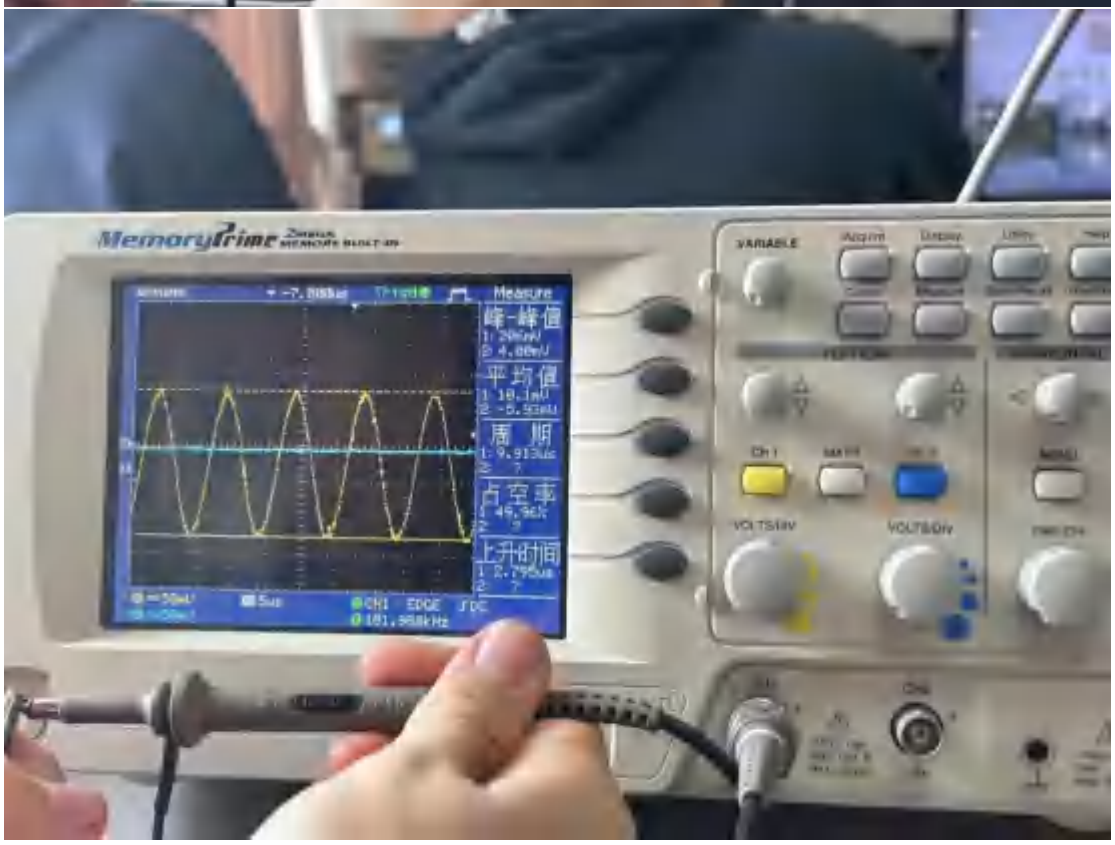
3. 示波器的使用：**10kHz** 的正弦波

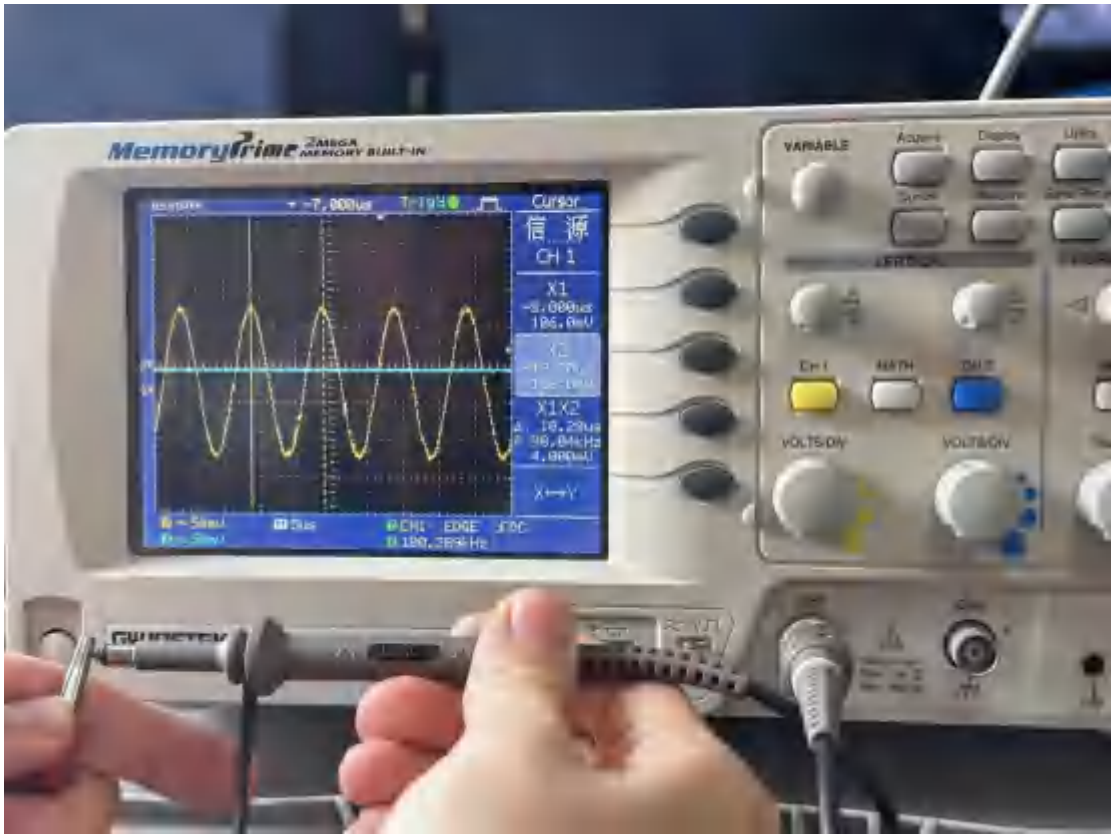




4. 频率 **100Hz** 的正弦波

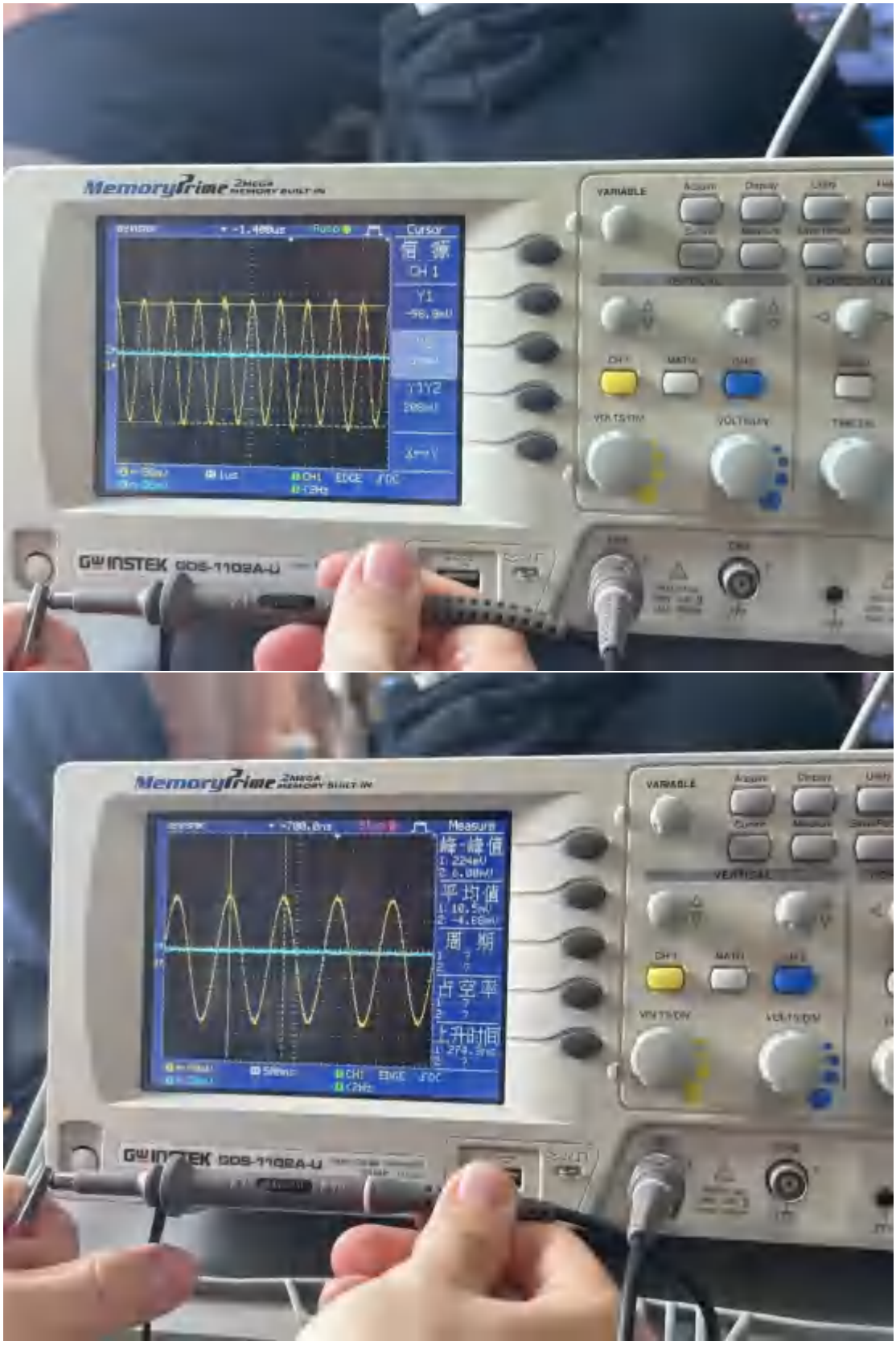

MemoryPrime
VARIABLE
Cursor
信源
CH 1
Y2
112mV
V1Y2
202mV
X↔Y
CH1 EDGE
101.88kHz


MemoryPrime
VARIABLE
Measure
峰-峰值
1: 200mV
2: 4.00mV
平均值
1: 10.1mV
2: -5.50mV
周 期
1: 9.915us
占空率
1: 49.9%
上升时间
1: 2.795us
CH1 EDGE
101.98kHz


MemoryPrime
VARIABLE
Cursor
信 源
CH 1
X1
X1X2
CH1
EDGE
GW INSTEK

5.1Mhz


MemoryPrime
VARIABLE
Cursor
信 源
CH 1
X1
X1X2
CH1
EDGE
GW INSTEK


GW INSTEK GDS-1102A-U
Cursor
CH 1
Measure
周 期
占空率
GW INSTEK GDS-1102A-U

5. 10kHz 的方波

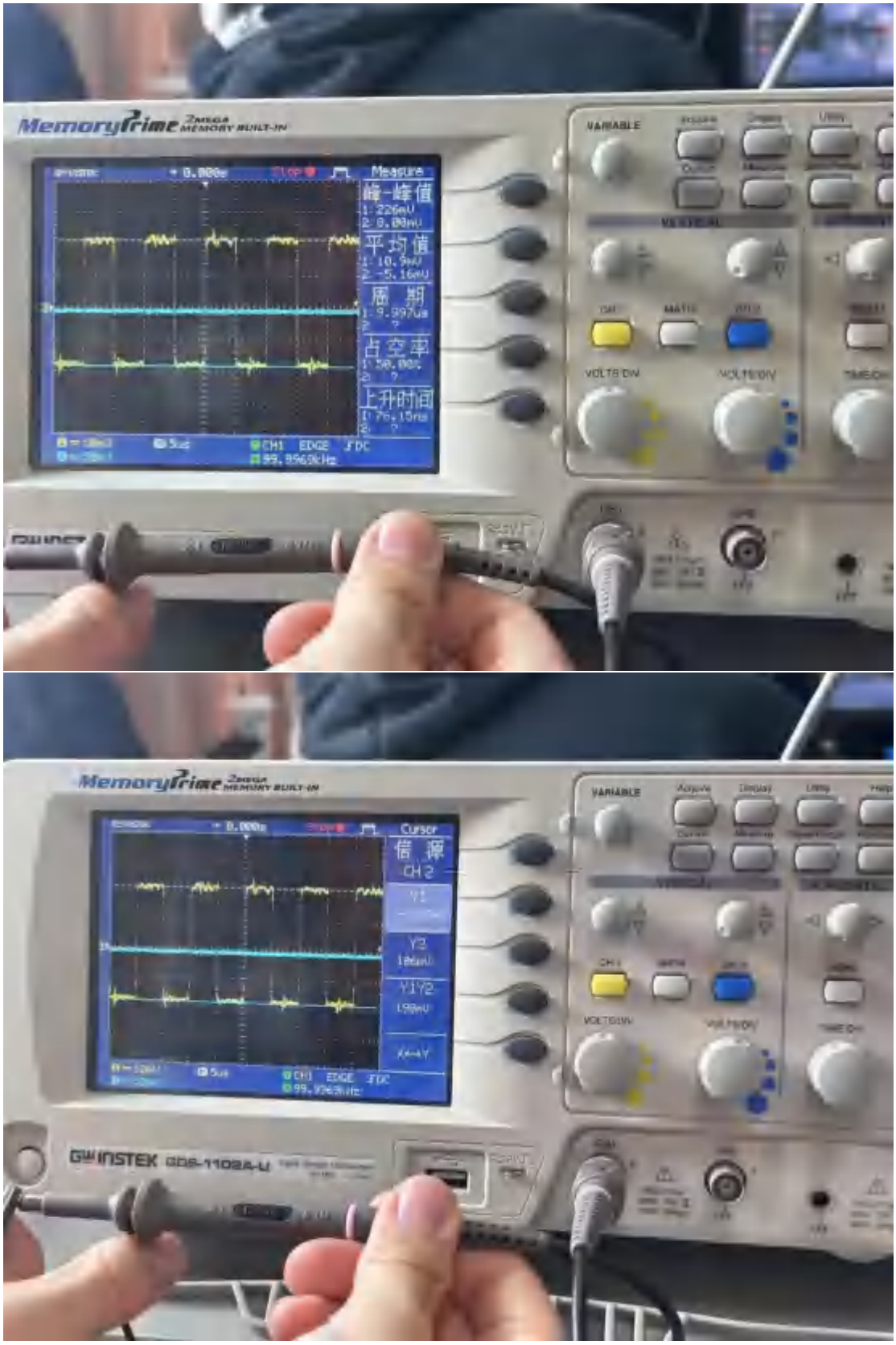

MemoryPrime
VARIABLE
Cursor
CH 1
X2
X1X2
GW INSTEK GDS-1102A-U


MemoryPrime
VARIABLE
Cursor
CH 2
Y1
Y1Y2
GW INSTEK GDS-1102A-U

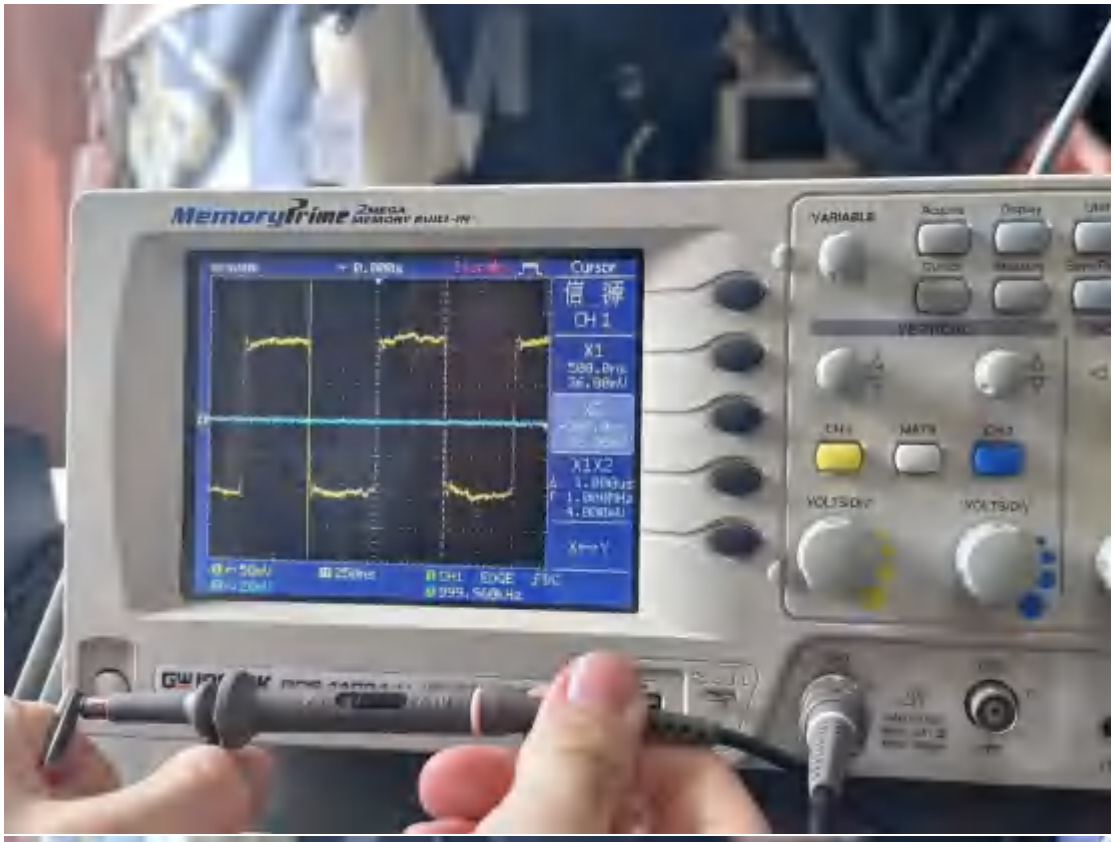

Measure
峰-峰值
平均值
周 期
占空率
上升时间
GW INSTEK GDS-1102A-U

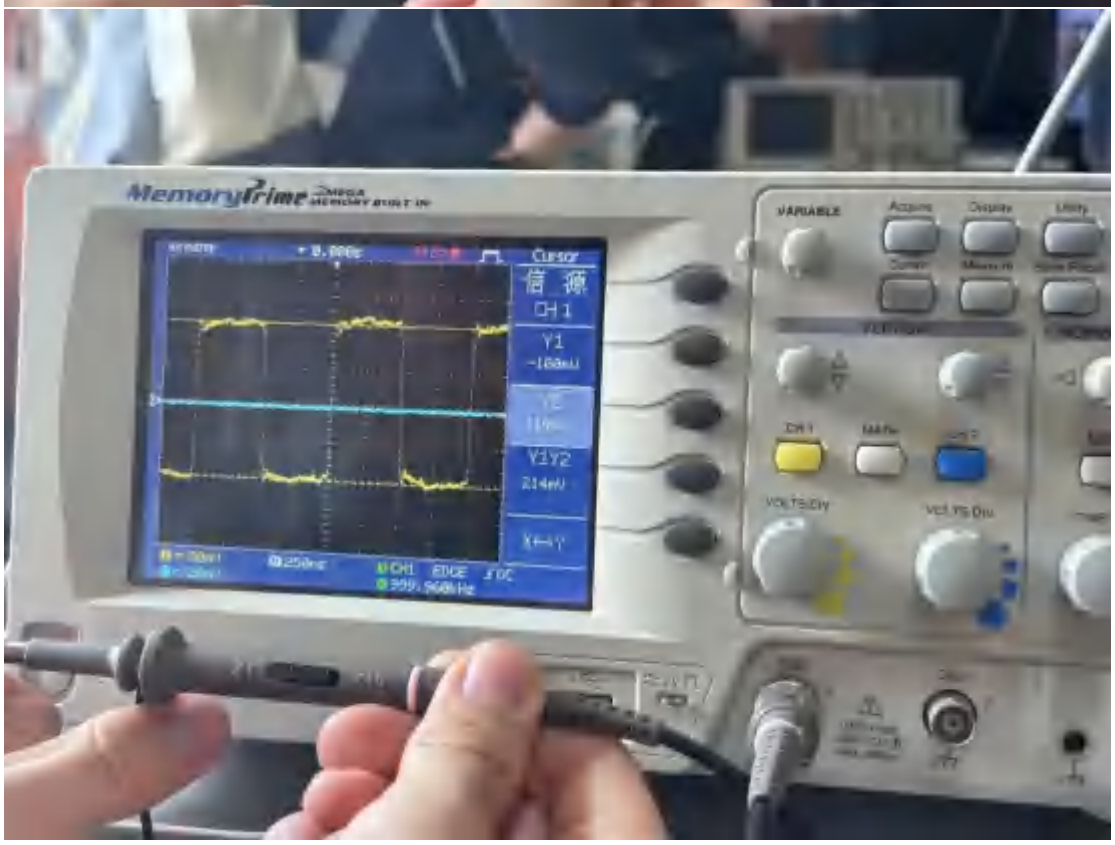
6.100Hz 的方波

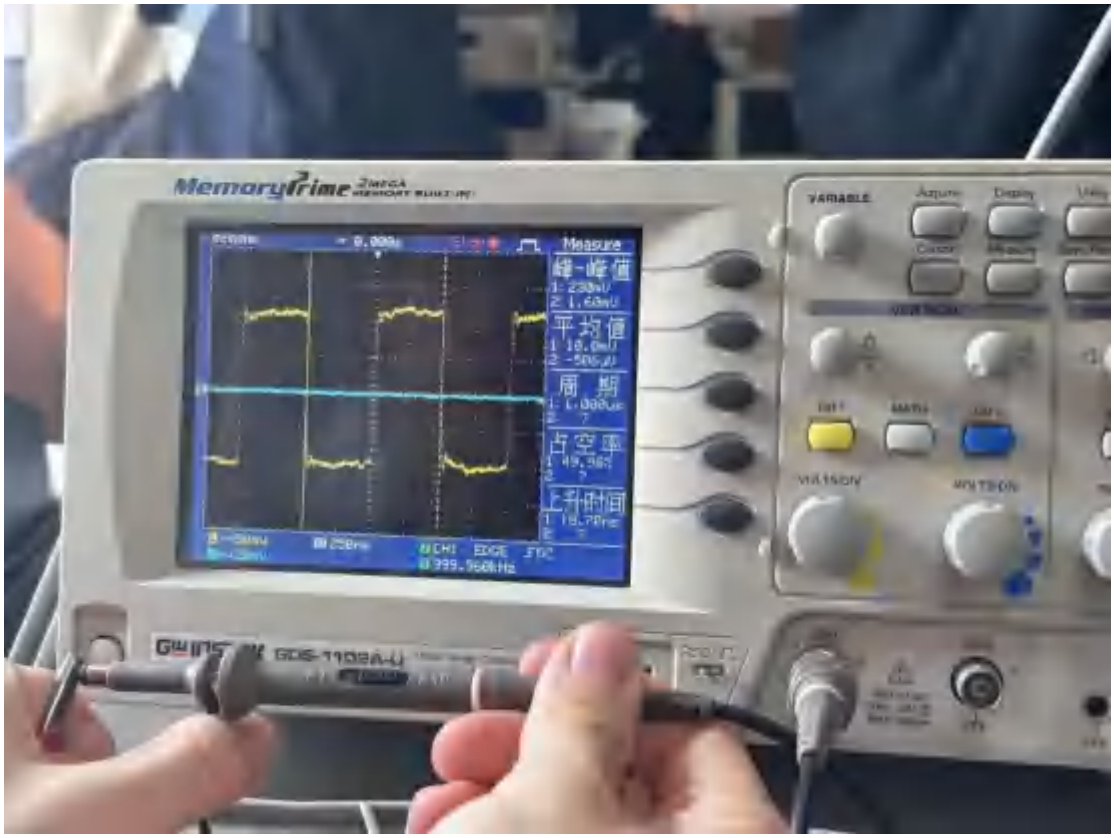

Cursor
CH2
GW INSTEK GDS-1102A-U


MemoryPrime
VARIABLE
Measure
峰-峰值
平均值
周期
占空率
上升时间
CH1 EDGE
MemoryPrime
VARIABLE
Cursor
信源
CH2
Y2
Y1Y2
GW INSTEK GDS-1102A-U

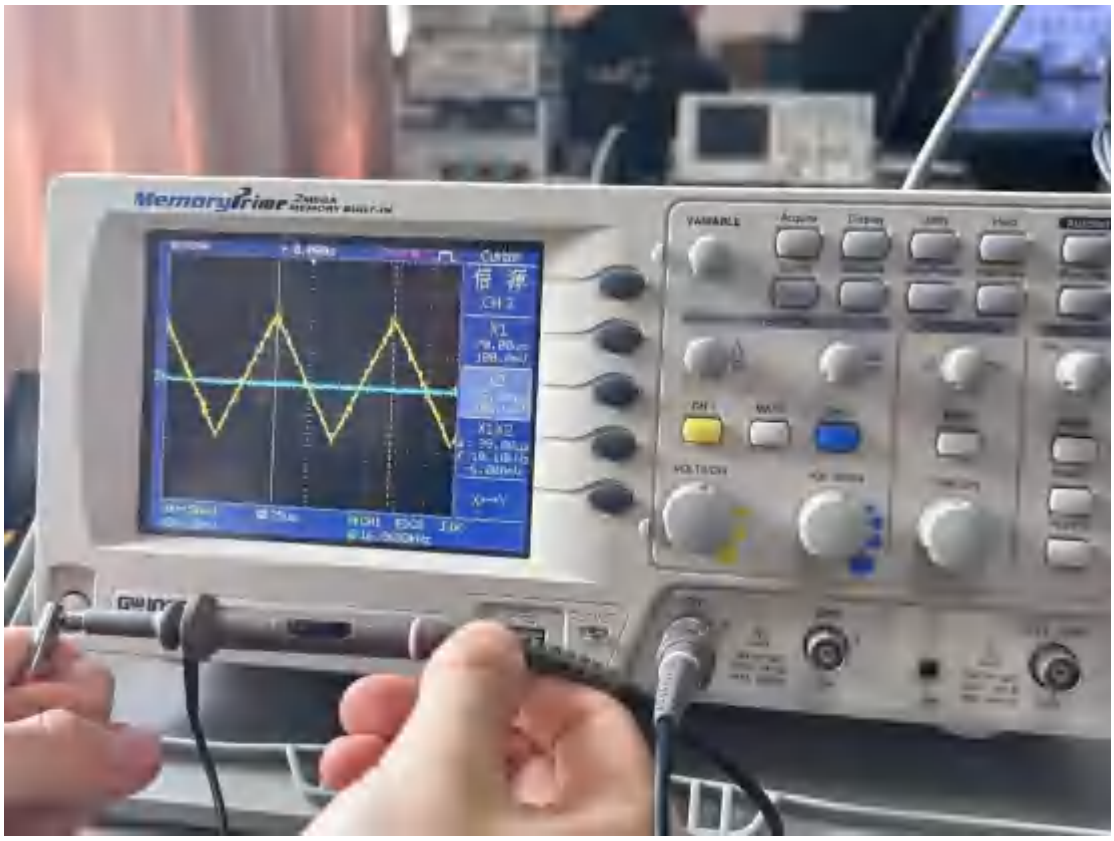
7.100kHz 的方波

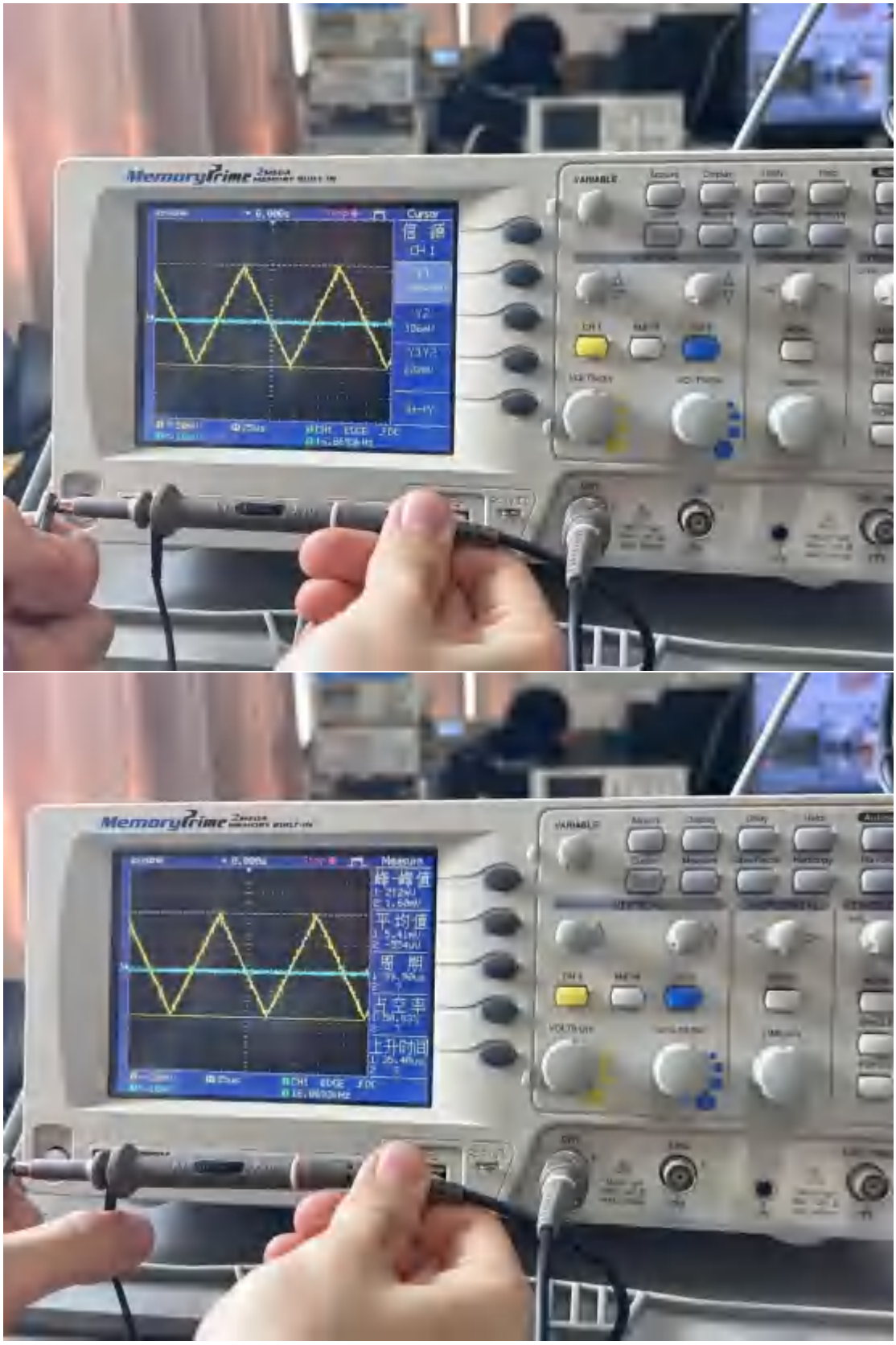

MemoryPrime
VARIABLE
Cursor
CH 1
X1
X1Y2
X-Y
CH1
MATH
CH2
VOLTS/DIV


MemoryPrime
VARIABLE
Acquire
Display
Utility
Cursor
Measure
Save/Recall
Cursor
CH 1
Y1
Y1Y2
X-Y
CH1
MATH
CH2
VOLTS/DIV
VOLTS/DIV

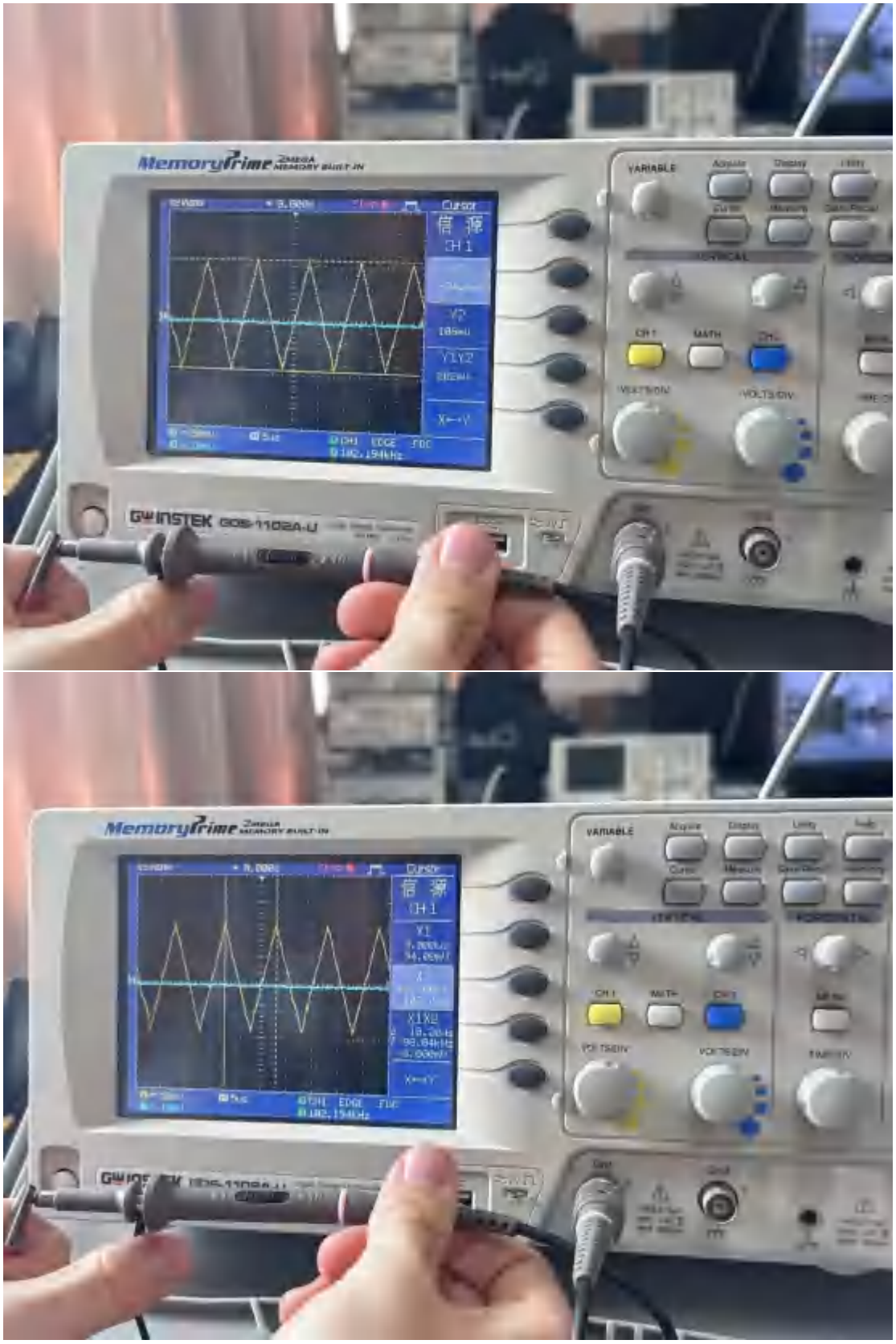


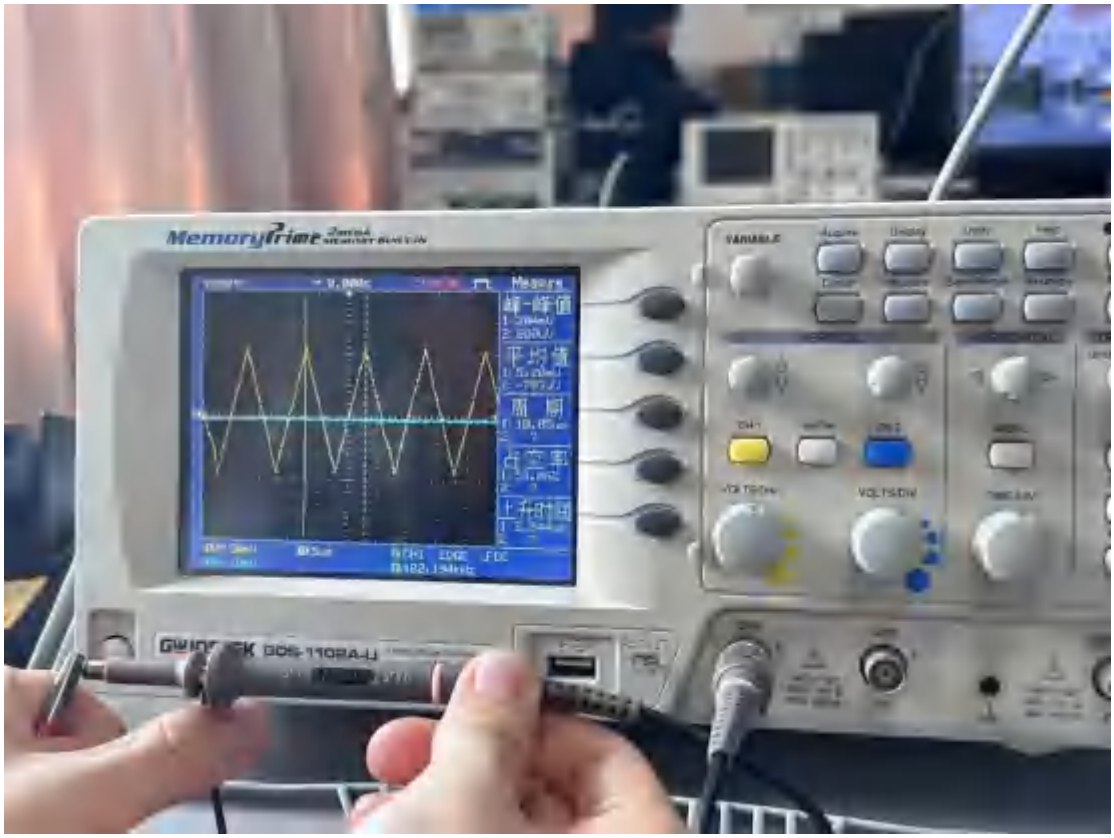
8.10Hz 的三角波



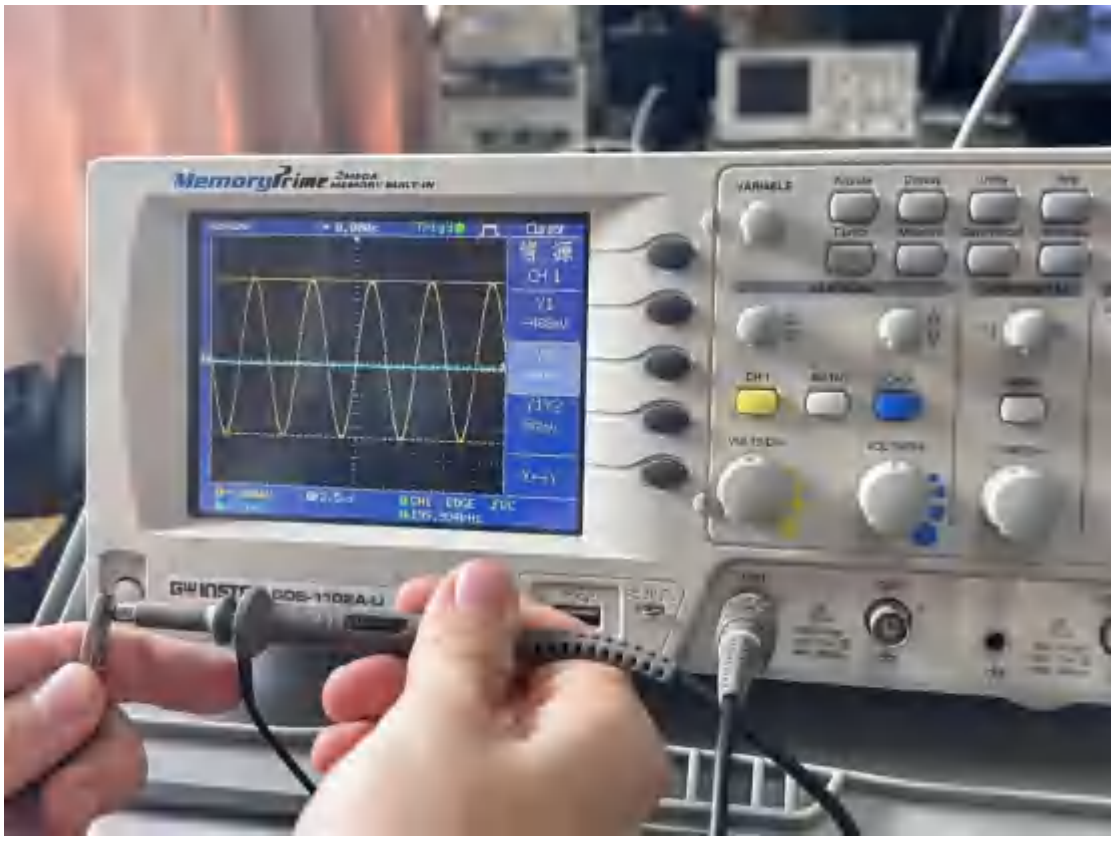

MemoryPrime
Cursor
MemoryPrime
Measure
峰-峰值
平均值
周期
占空率
上升时间

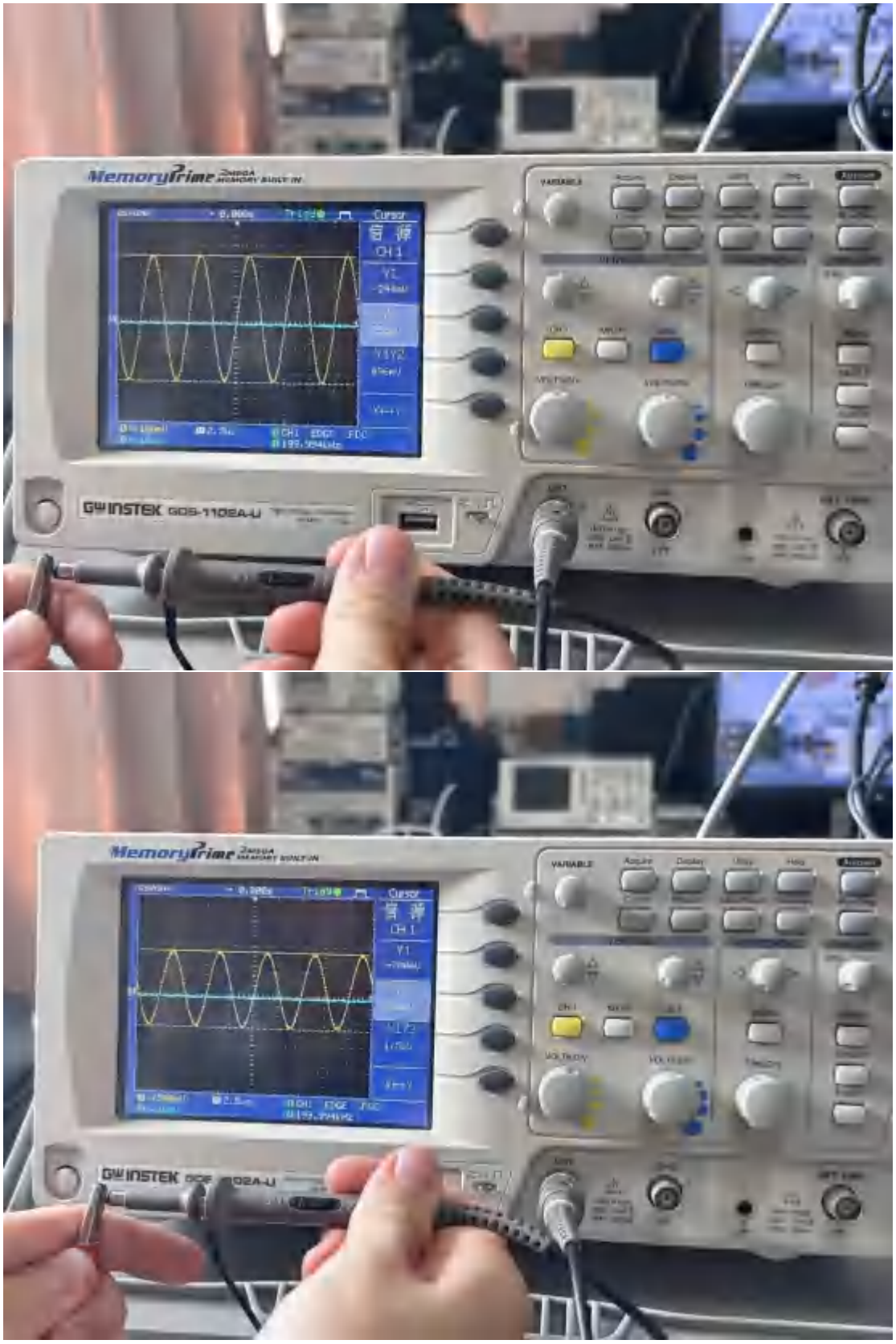
9.100Hz 的三角波

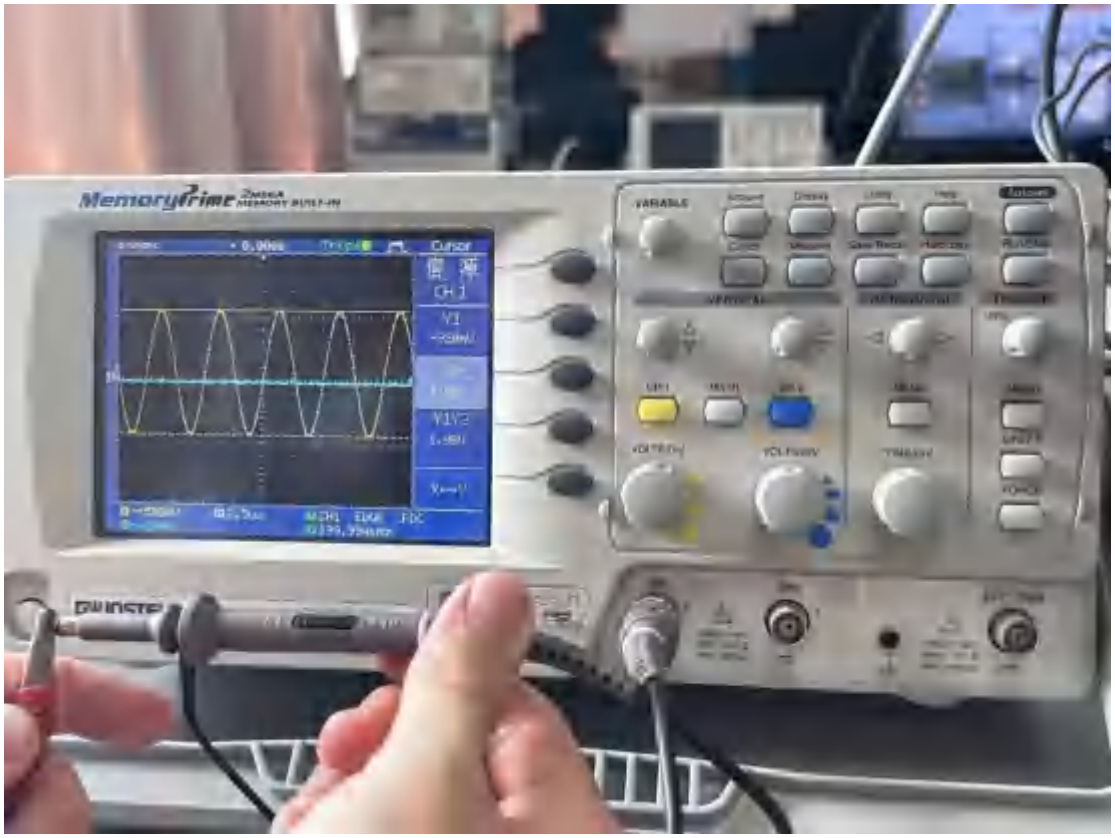

MemoryPrime
VARIABLE
Cursor
CH 1
GW INSTEK GDS-1102A-U



10.信号频率保持 **200KHz** 不变 改变幅度 理论幅度为 **0.5 1 1.5 2.0** 四个




MemoryPrime
GW INSTEK GDS-1102A-U
Cursor
CH1


MemoryPrime
VARIABLE
Cursor
CH1